

Coaction of Mellin Motives

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Periods

Definition: A **period** is:

$$\int_{\sigma} \omega \in \mathcal{P},$$

for X an algebraic variety, ω an algebraic differential form on X , σ a cycle on $X(\mathbb{C})$.

Conjecture: There exists an algebraic group $\mathrm{Gal}^{\mathrm{mot}}$ over $\mathbb{Q}$

$$\mathrm{Gal}^{\mathrm{mot}} \times \mathcal{P} \rightarrow \mathcal{P}$$

preserving *all* algebraic relations amongst periods.

What are the Motivic Periods of π_1 ?

- To avoid conjecture, we use motivic periods.
- Necessary data: Betti and de Rham realisations of motivic π_1 .

Definition: A **motivic period** is the symbol

$$[\mathcal{O}(\pi_1^{\text{mot}}), \omega, \sigma]^{\text{mot}} \in \mathcal{P}^{\text{mot}}, \quad \omega \in \mathcal{O}(\pi_1^{\text{dR}}), \quad \sigma \in \mathcal{O}(\pi_1^{\text{B}})^{\vee},$$

modulo relations: Stokes, bilinearity, change of variables, etc.

Question: How to construct the **period map**?

$$\text{per} : \mathcal{P}^{\text{mot}} \rightarrow \mathbb{C} \quad : \quad [\omega, \sigma]^{\text{mot}} \mapsto \langle \omega, \text{comparison}_{\text{B,dR}}(\sigma) \rangle?$$

Answer: Iterated Integrals

Definition: An **iterated integral** is:

$$\int_{\sigma} \omega_1 \dots \omega_r := \int_{0 \leq t_1 \leq \dots \leq t_r \leq 1} \sigma^* \omega_1(t_1) \dots \sigma^* \omega_r(t_r).$$

Theorem (Chen 1977): There is an **iterated integration pairing**:

$$\mathcal{O}(\pi_1^{\text{dR}}) \cong \mathcal{O}(\pi_1^B)^{\vee} \quad : \quad \omega_1 \dots \omega_r \mapsto \left(\sigma \mapsto \int_{\sigma} \omega_1 \dots \omega_r \right).$$

where $\mathcal{O}(\pi_1^B)$ is the prounipotent completion of $\pi_1^{\text{topological}}$; and $\mathcal{O}(\pi_1^{\text{dR}})$ is the reduced bar complex of 1-forms.

E.g. Motivic Periods of $\pi_1^{\text{mot}}(\mathbb{P}^1 \setminus \{0, 1, \infty\})$

- **Betti:** Image of the ‘droit chemin’ and a loop under inclusion $\pi_1^{\text{top}} \hookrightarrow \pi_1^B(\mathbb{Q})$:

$$\begin{array}{ccc}
 \begin{array}{c} \xrightarrow{1_0} \\ 0 \bullet \end{array} & \xrightarrow{\text{dch}} & \begin{array}{c} \xrightarrow{-1_1} \\ \bullet 1 \end{array} \mapsto {}_0 1_1^B, \\
 & & \gamma \left(\begin{array}{c} \xrightarrow{1_0} \\ 0 \bullet \\ \xrightarrow{-1_0} \end{array} \right) \mapsto {}_0 1_0^B.
 \end{array}$$

- **de Rham:** Words written with letters $e_a := d \log(x - a)$, with $a \in \{0, 1\}$.
- **Motivic periods:** $\mathbb{L}^{\text{mot}} := [e_0, {}_0 1_0^B]^{\text{mot}}$, $\zeta^{\text{mot}}(k) := [e_1 e_0^{k-1}, {}_0 1_1^B]^{\text{mot}}$.

$$\text{per } \mathbb{L}^{\text{mot}} = \int_{\gamma} d \log(x) = 2\pi i,$$

$$\text{per } \zeta^{\text{mot}}(k) = \int_{\text{dch}} d \log(x - 1) d \log(x) \dots d \log(x) = \sum_{n > 0} \frac{1}{n^k} = \zeta(k).$$

Motivic Galois Group

- (*Belyi 1979, Grothendieck 1984, Ihara 1990*): The absolute Galois group acts faithfully on the completed fundamental group:

$$\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \simeq \widehat{\pi_1}(\mathbb{P}^1 \setminus \{0, 1, \infty\})$$

↓

- (*Deligne 1989, Brown 2014*): The (de Rham realisation of the) **motivic** Galois group acts faithfully on the (de Rham realisation of the) **motivic** fundamental group:

$$\mathrm{Gal}^{\mathrm{dR}} \simeq \pi_1^{\mathrm{dR}}(\mathbb{P}^1 \setminus \{0, 1, \infty\})$$

Motivic Galois (Co)Action

- In practice, it is easier to write *coaction* than action.

$$\begin{aligned} \mathrm{Gal}^{\mathrm{dR}} \times \mathcal{P}^{\mathrm{mot}} &\rightarrow \mathcal{P}^{\mathrm{mot}} \\ (g, [\omega, \sigma]^{\mathrm{mot}}) &\mapsto [g.\omega, \sigma]^{\mathrm{mot}} \end{aligned}$$

$$\begin{aligned} &\xleftrightarrow[\text{dualise}]{} \mathcal{P}^{\mathrm{mot}} \rightarrow \mathcal{P}^{\mathrm{mot}} \otimes \mathcal{O}(\mathrm{Gal}^{\mathrm{dR}}) \\ [\omega, \sigma]^{\mathrm{mot}} &\mapsto \sum_{\text{basis } e_i \in \mathcal{O}(\pi_1^{\mathrm{dR}})} [e_i, \sigma]^{\mathrm{mot}} \otimes [\omega, e_i^\vee]^{\mathrm{dR}} \end{aligned}$$

Example:

$$g.\mathbb{L}^{\mathrm{mot}} = \lambda(g)\mathbb{L}^{\mathrm{mot}},$$

$$\Delta\mathbb{L}^{\mathrm{mot}} = \mathbb{L}^{\mathrm{mot}} \otimes \mathbb{L}^{\mathrm{dR}}$$

$$g.\zeta^{\mathrm{mot}}(k) = \lambda(g)^k \zeta^{\mathrm{mot}}(k) + a^{(k)}(g)$$

$$\Delta\zeta^{\mathrm{mot}}(k) = \zeta^{\mathrm{mot}}(k) \otimes \lambda^k + 1 \otimes a^{(k)},$$

$$\text{where } \lambda(g), a^{(k)}(g) \in \mathbb{Q}.$$

$$\begin{aligned} &\text{where } \lambda, a^{(k)} \in \mathcal{O}(\mathrm{Gal}^{\mathrm{dR}}), \text{ and} \\ &\mathbb{L}^{\mathrm{dR}} := [e_0, e_0^\vee]. \end{aligned}$$

Motivic Galois Coaction on $\left[e_{a_1} \dots e_{a_N}, a_0 1_{a_{N+1}}^B \right]^{\text{mot}}$

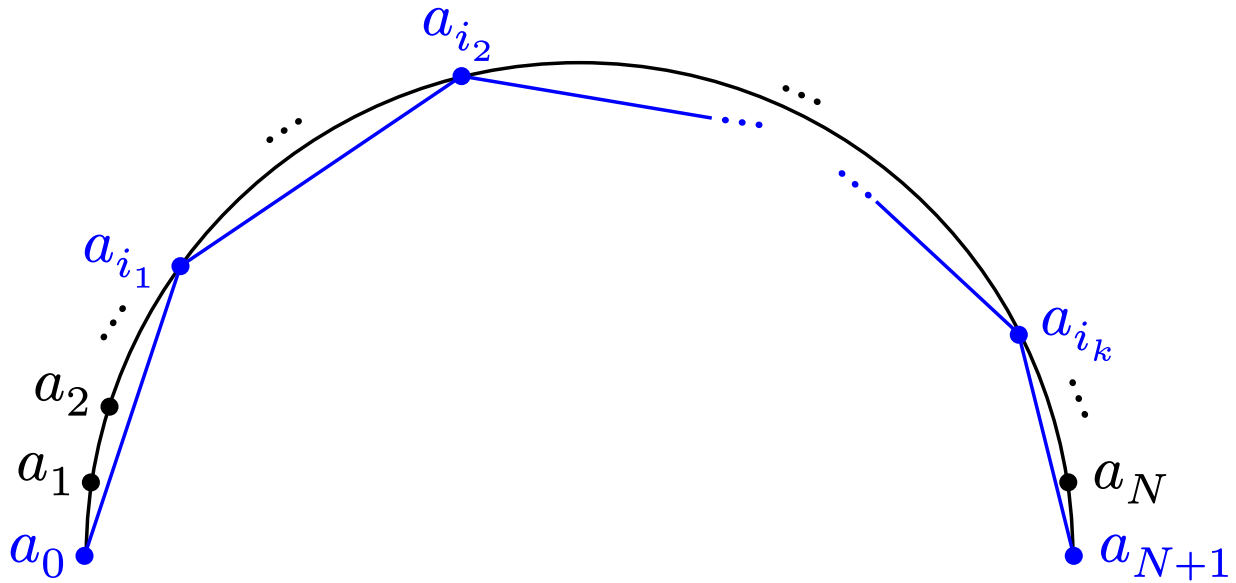


Figure 1:

$$\sum_{0=i_0 < i_1 < \dots < i_k < i_{k+1}=N+1} \prod_{p=0}^k \left[a_{i_p+1}, \dots, a_{i_{p+1}-1}; a_{i_p} 1_{a_{i_{p+1}}}^B \right]^{\text{mot}} \otimes \left[a_{i_1}, \dots, a_{i_k} a_0 1_{a_{n+1}}^{\text{dR}} \right]^{\text{dR}}$$

Coaction under Dimensional Regularisation

- In physical applications, **regularisation** of integrals is paramount.

Conjecture (ABDGM 2019): Dimensional regularisation is compatible with (diagrammatic) coaction.

Definition: A general **Mellin transform** is $\mathcal{M}(s)\left\{\int_{\sigma}\omega\right\}:=\int_{\sigma}f^s\omega$, for f algebraic.

Question: Can we calculate the coaction of the Taylor expansion:

$$\mathcal{M}(s)\left\{\int_{\sigma}\omega\right\}=\sum_{n\geq 0}\frac{s^n}{n!}\int_{\sigma}\log^n(f)\omega?$$

E.g. Motivic Beta Function

Definition: The **Euler beta function** is the period $\beta(s, t) := \int_0^1 x^s (1-x)^t \frac{dx}{x(1-x)}$.

- Lift the famous Taylor expansion to a **Mellin** motivic period:

$$\beta(s, t) = \frac{s+t}{st} \exp \left(\sum_{n \geq 2} \frac{(-1)^n}{n} \zeta(n) (s^n + t^n - (s+t)^n) \right)$$

$\downarrow$

$$\beta^{\text{mot}}(s, t) := \frac{s+t}{st} \exp \left(\sum_{n \geq 2} \frac{(-1)^n}{n} \zeta^{\text{mot}}(n) (s^n + t^n - (s+t)^n) \right) \in \mathcal{P}^{\text{mot}}[[s, t]]$$

E.g. Motivic Beta Function Coaction

- Apply Δ termwise to the motivic period:

$$\begin{aligned}\Rightarrow \Delta \beta^{\text{mot}}(s, t) &= \beta^{\text{mot}}(s, t) \otimes \frac{st}{s+t} \exp \left(\sum_{n \geq 2} \frac{(-1)^n}{n} \frac{a^{(n)}}{\lambda^n} (s^n + t^n - (s+t)^n) \right) \\ &\in \mathcal{P}^{\text{mot}}[[s, t]] \otimes \mathcal{O}(\text{Gal}^{\text{dR}})[[s, t]]\end{aligned}$$

- **Current work:** systematise this ‘lifting’ construction of Mellin motives, and obtain an explicit coaction formula.

Thank you!



A melon motif. Béla Iványi-Grünwald, Autumn Still-life (1938).